

MagMat-Discovery

Physics-Informed Machine Learning and Screening for Magnetic Materials

System Architecture, Methodology & Model Validation

Technical Report & Model Guide

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Web Platform: Interactive Streamlit Discovery Engine

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Executive Summary

Discovering high-performance permanent magnets with reduced rare-earth content is important for electric vehicles, wind turbines, and industrial machinery. Developing sustainable alternatives requires identifying materials that satisfy three physical criteria: elevated Curie temperature ($T_C > 400$ K), substantial saturation magnetization ($\mu_0 M_s > 1.0$ T), and high coercivity ($H_c > 800$ kA/m). However, machine learning across magnetic materials is hindered by the physical disconnect between exchange-driven intrinsic properties and microstructure-sensitive extrinsic properties, data leakage across near-identical dopants and polymorphs, and the risk of unphysical predictions.

MagMat-Discovery addresses these challenges through four primary objectives:

1. **Unified Multi-Target Modeling:** Simultaneously predict 8 intrinsic and extrinsic magnetic properties spanning ground-state phase, transition temperatures, and hysteresis behavior.
2. **Leak-Free Grouped Evaluation:** Validate models using 5-fold cross-validation partitioned strictly by reduced chemical formula, ensuring identical compositions and polymorphs never cross train–test splits.
3. **Physics-Enforced Cascades:** Connect intrinsic predictions (K_1, M_s) to extrinsic properties ($H_c, (BH)_{\max}$) using micromagnetic relationships and thermodynamic bounds.
4. **Calibrated Uncertainty and Screening:** Provide 90% conformal prediction intervals for every continuous target and deploy an interactive web platform for candidate discovery.

By combining curated experimental records, domain physics features, a custom stoichiometric graph transformer (**MagFormer**), specialist ensembles, and conformal calibration, the platform delivers strong predictive fidelity for magnetic phase and ordering temperatures, alongside physically bounded rankings for extrinsic hysteresis behavior.

Keywords: Magnetic Materials, Physics-Informed Machine Learning, Graph Transformers, Conformal Calibration, Coercivity, Permanent Magnets, Materials Screening.

Glossary of Key Terms

Table 1 summarizes the primary magnetic properties and machine learning terminology used throughout this report.

Table 1: Glossary of magnetic properties, physical units, and computational terms.

Term	Typical Unit	Physical Meaning / Definition
T_C (Curie Temp.)	K	Temperature above which a ferromagnetic material loses its spontaneous magnetization.
T_N (Néel Temp.)	K	Temperature above which an antiferromagnetic material becomes paramagnetic.
M_s (Saturation Mag.)	emu/g or T	Maximum magnetization achieved in an applied magnetic field. Reported in mass units (emu/g) or magnetic polarization $\mu_0 M_s$ (T).
H_c (Coercivity)	A/m or kA/m	Applied magnetic field required to reduce magnetization back to zero after saturation.
K_1 (Anisotropy Constant)	J/m ³	Energy density required to rotate magnetization away from the preferred crystal axis.
$(BH)_{\max}$ (Energy Product)	kJ/m ³	Primary figure of merit for permanent magnets, representing stored magnetic energy density.
θ_p (Paramagnetic Curie Temp.)	K	High-temperature intercept from Curie-Weiss susceptibility fitting; its sign reflects dominant exchange coupling.
FM / AFM / NM	Discrete	Ground-state magnetic ordering: Ferromagnetic (parallel), Antiferromagnetic (antiparallel), or Non-magnetic.
Out-of-Fold (OOF)	–	Predictions made on data held out during cross-validation, ensuring unbiased inputs to downstream models.
Conformal Range	Physical	A calibrated uncertainty interval constructed from validation residuals, expected to contain the true value approximately 90% of the time.
dex	log ₁₀	Decimal exponent (order of magnitude). An error of 0.65 dex means a factor of $10^{0.65} \approx 4.5$.

1 Data and Partitioning

1.1 Data Sources

The platform aggregates experimental measurements with crystallographic symmetry databases to establish a reliable foundation for model training and evaluation:

- **NEMAD Database:** 51,394 curated experimental records from the Non-Equilibrium Magnetic Materials Database [1], providing physical ground truths for magnetic phase, T_C , T_N , M_s , K_1 , H_c , $(BH)_{\max}$, and θ_p .
- **Materials Project (U.S. Department of Energy / DOE):** DFT crystallographic symmetries (space groups 1–230, point groups, crystal systems, and densities) from the open repository developed under the U.S. DOE Office of Science [2]. Computed ground-state energies and DFT magnetic moments are excluded to prevent target leakage when evaluating uncharacterized materials.
- **Bilbao MAGNDATA:** Magnetic space group ground truths and Shubnikov symmetry classifications for magnetic phase cross-referencing.

1.2 Data Cleaning and Grouped Splits

Raw records are sanitized by canonicalizing chemical formulas to coprime integer stoichiometries using `pymatgen`. Properties spanning multiple orders of magnitude (H_c , K_1 , $(BH)_{\max}$) are log-transformed to stabilize variance during training.

To ensure realistic generalization estimates, cross-validation is performed using a 5-fold `GroupKFold` protocol partitioned strictly by `reduced_formula`. Under this partitioning, identical chemical compositions and pure polymorphs (such as different crystal phases of Fe_3Pt or repeated entries of $\text{Nd}_2\text{Fe}_{14}\text{B}$) are isolated entirely within a single fold. This guarantees that test folds contain only unseen compounds, evaluating whether models generalize across chemical families rather than memorizing known parent stoichiometries.

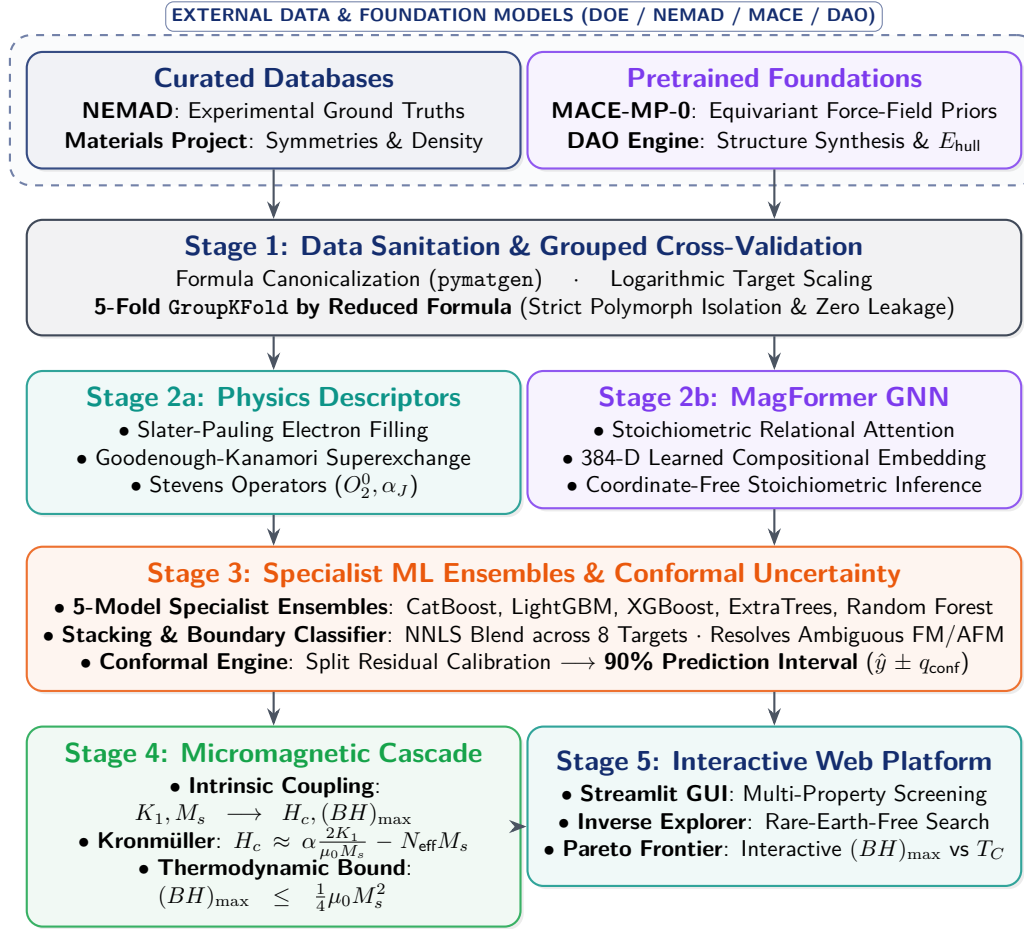


Figure 1: System architecture and end-to-end data flow of MagMat-Discovery. Curated databases (NEMAD, Materials Project funded by DOE) and pretrained foundation models (MACE-MP-0, DAO) provide baseline ground truths and structural embeddings. In-house pipeline components (MagFormer stoichiometric transformer, multi-tier physics featurizers, 5-model specialist ensembles, micromagnetic physical cascades, and conformal uncertainty calibration) perform leak-free training and screening.

2 Methodology

2.1 Architecture and Provenance

Figure 1 shows the system architecture of MagMat-Discovery. A central design principle is clear provenance: distinguishing imported external foundation models and government databases from in-house algorithms (Table 2). External resources supply experimental ground truths, crystal symmetries, equivariant force-field priors, and generative crystal structures. All property regressors, boundary classifiers, micromagnetic cascades, and conformal calibrators were developed, trained, and validated in-house.

Table 2 provides a detailed accounting of data and model origins across all pipeline stages.

2.2 Representations and Structure Synthesis

To capture both fundamental chemistry and crystal physics, the pipeline employs a dual representation approach:

1. **Multi-Tier Physics Descriptors:** Features are structured into four hierarchical physical

Table 2: Provenance of databases, foundation models, and in-house algorithmic components in MagMat-Discovery.

Component	Category	Origin / Training Source	Exact Role in MagMat-Discovery
Materials Project (U.S. DOE)	External DFT Database	High-throughput DFT funded by U.S. DOE Office of Science [2]	Supplies crystallographic symmetries (space groups 1–230), point groups, crystal systems, and densities. DFT total energies and magnetic moments are excluded to prevent target leakage.
NEMAD Database	External Experimental Database	Curated experimental literature compilation [1]	Provides ground-truth experimental targets ($T_C, T_N, M_s, H_c, K_1, (BH)_{\max}$, phase) for 5-fold grouped model training and benchmark validation.
MACE-MP-0 Foundation GNN	External Pretrained Foundation Model	Equivariant neural network pretrained on DOE MP trajectories [3]	Extracts higher-order equivariant atomic cluster expansion descriptors, cohesive energies, and stress tensors for compounds with known 3D crystal structures.
DAO Crystal Foundation Engine	External Pretrained Foundation Model	Diffusion crystal foundation model (Huang et al., <i>Nat. Commun.</i> 2026) [4]	Employs DAO-G to synthesize 3D periodic crystal structures from stoichiometry and DAO-P to calculate thermodynamic distance to convex hull (E_{hull} in eV/atom).
MagFormer Stoichiometric GNN	In-House Architecture (Trained from Scratch)	Designed and trained in this pipeline on chemical compositions	Stoichiometric self-attention graph transformer that embeds chemical formulas into 384-D representation vectors without requiring atomic coordinates.
Multi-Tier Physics Descriptors	In-House Domain Feature Engineering	Derived from solid-state and exchange physics theory	Handcrafted features encoding Slater-Pauling valence filling ($\text{VEC} \approx 8.3$), Goodenough-Kanamori superexchange proxies, spin-orbit terms, and Stevens operators (O_2^0, α_J).
5-Model Specialist Ensembles	In-House Machine Learning Models	Trained on leak-free 5-fold grouped splits in this work	Diverse ensemble (CatBoost, LightGBM, XGBoost, ExtraTrees, Random Forest) with non-negative least squares meta-blending for all 8 target properties.
Boundary Phase Classifier	In-House Machine Learning Model	Trained on ambiguous boundary compounds in this work	Secondary classifier resolving close ferromagnetic vs. antiferromagnetic ground states ($ P(\text{FM}) - P(\text{AFM}) < 0.20$).
Micromagnetic Physical Cascade	In-House Physical Post-Processing	Formulated from micromagnetic theory [5]	Enforces the Kronmüller coercivity relationship and caps maximum energy product at the thermodynamic limit $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$.
Conformal Uncertainty Engine	In-House Uncertainty Calibration	Split conformal prediction on out-of-fold residuals [6]	Computes rigorous distribution-free 90% uncertainty intervals ($y \pm q_{\text{conf}}$) calibrated specifically to out-of-fold validation errors.

tiers (formal equations in Appendix A):

- *Tier 1 (Stoichiometry)*: Elemental mass fractions, atomic numbers, covalent radii, and Pauling electronegativities.
- *Tier 2 (Electronic Structure)*: Valence electron concentration (VEC) relative to the Slater-Pauling maximum ($\text{VEC} \approx 8.3$), Goodenough-Kanamori superexchange proxies, and spin-orbit scaling ($\propto Z^4$).
- *Tier 3 (Crystal Field)*: Crystal system indicators and second-order Stevens operator terms (O_2^0, α_J) derived from Materials Project symmetries.
- *Tier 4 (Equivariant Embeddings)*: Equivariant atomic representations extracted from MACE-MP-0 [3] when crystal coordinates are known.

2. MagFormer Graph Transformer: When screening uncharacterized alloys lacking crystal coordinates, **MagFormer** encodes chemical formulas directly into 384-dimensional representation vectors using relational self-attention over constituent elements (Appendix B).

3. DAO Structure Synthesis and Stability: For candidates requiring 3D structural validation, the platform interfaces with the DAO diffusion foundation model [4]. DAO-G synthesizes 3D unit cells from formula strings, while DAO-P evaluates energy above the convex hull (E_{hull}). Compositions with $E_{\text{hull}} < 0.05$ eV/atom are classified as thermodynamically

synthesizable.

2.3 Ensembles, Cascades, and Uncertainty

Each target is modeled using an ensemble of five diverse regressors: CatBoost, LightGBM, XGBoost, ExtraTrees, and Random Forest, combined via non-negative least squares (NNLS) meta-blending. A secondary classifier resolves close FM/AFM phase boundaries ($|P(\text{FM}) - P(\text{AFM})| < 0.20$).

To prevent unphysical predictions in extrinsic hysteresis targets, the pipeline executes a physical cascade:

1. Intrinsic properties (K_1 and M_s) are predicted first.
2. Out-of-fold estimates are supplied to the coercivity (H_c) and energy product ($(BH)_{\max}$) models, grounding coercivity in the Kronmüller relationship [5]:

$$H_c \approx \alpha_{\text{micro}} \frac{2K_1}{\mu_0 M_s} - N_{\text{eff}} M_s \quad (1)$$

where α_{micro} accounts for microstructural defects and N_{eff} is the effective demagnetizing factor.

3. Predicted energy products are bounded by the thermodynamic limit: $(BH)_{\max} \leq \frac{1}{4} \mu_0 M_s^2$.

Every continuous prediction includes a distribution-free uncertainty range via split conformal prediction [6]. The 90th percentile of absolute out-of-fold residuals defines a calibration margin:

$$q_{\text{conf}} = \text{Quantile}_{0.90}(\{|y_i - \hat{y}_i|\}_{i=1}^n) \quad (2)$$

yielding calibrated prediction intervals $[\hat{y} - q_{\text{conf}}, \hat{y} + q_{\text{conf}}]$ with guaranteed 90% coverage.

Table 3: Comprehensive 5-fold GroupKFold cross-validation results across all 8 magnetic targets.

#	Target Property	Unit	Scale	Count n	Primary Metric	Mean Error	Predictive Utility
1	Ground-State Phase	Discrete	Class	35,045	91.9% Accuracy	Macro F_1 : 0.908	Strong (Reliable phase filtering)
2	Curie Temperature (T_C)	K	Linear	15,518	$R^2 = 0.816$	MAE = 69.8 K	Strong (Thermal threshold filter)
3	Néel Temperature (T_N)	K	Linear	7,734	$R^2 = 0.741$	MAE = 48.6 K	Strong (Accurate AFM boundary)
4	Anisotropy Constant (K_1)	J/m ³	Symlog	3,938	$R^2 = 0.461$	MAE = 0.95 dex	Moderate (Easy-axis candidate ranking)
5	Coercivity (H_c)	A/m	log ₁₀	6,980	$R^2 = 0.622$	MAE = 0.65 dex	Moderate (Hard vs soft screening)
6	Saturation Magnetization (M_s)	emu/g	Linear	7,619	$R^2 = 0.539$	MAE = 26.1 emu/g	Moderate (Moment estimation)
7	Energy Product ($(BH)_{\max}$)	kJ/m ³	log ₁₀	1,379	$R^2 = 0.507$	MAE = 0.27 dex	Moderate (Bounded density estimate)
8	Paramagnetic Curie Temp. (θ_p)	K	Linear	5,474	$R^2 = 0.607$	MAE = 64.7 K	Moderate (Exchange coupling sign)

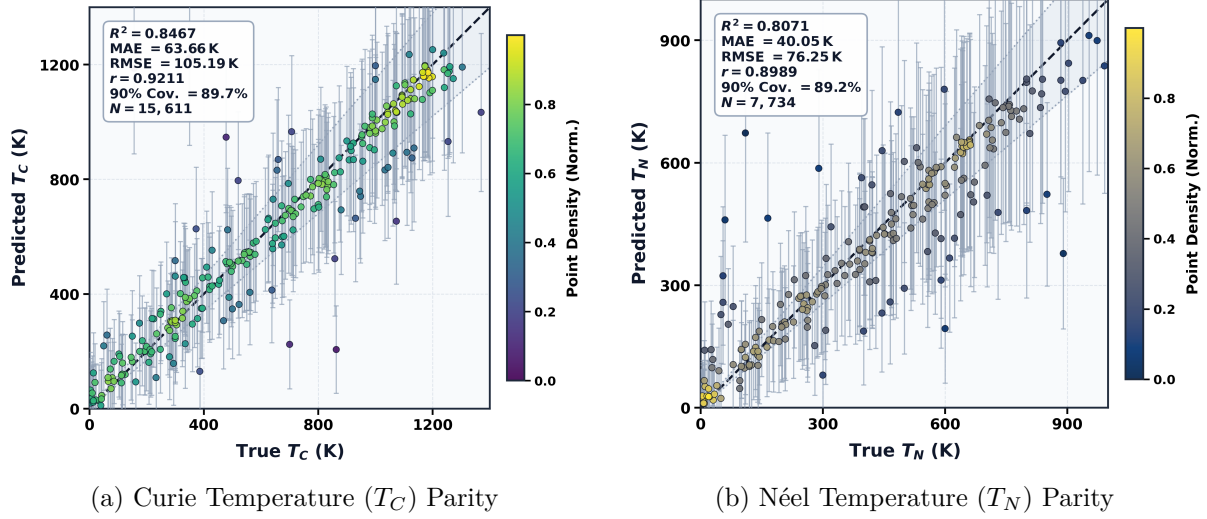


Figure 2: Parity plots for Curie (T_C) and Néel (T_N) transition temperatures. Points close to the diagonal line represent accurate predictions. Both models demonstrate strong fidelity across the operating temperature range ($R^2 = 0.82$ and 0.74), with scatter increasing slightly below 100 K.

3 Model Validation

3.1 Benchmark Performance

Model performance across the 8 magnetic targets is evaluated using 5-fold GroupKFold cross-validation (Section 1.2). Prediction quality is quantified by the coefficient of determination (R^2) and mean absolute error (MAE), reported in physical units for linear properties (K, emu/g) and in decimal orders of magnitude (dex) for wide-span logarithmic targets (H_c , K_1 , $(BH)_{\max}$). Table 3 summarizes the benchmark results across all targets.

3.2 Parity Analysis

Across the 8 targets, a clear physical distinction emerges between exchange-governed intrinsic properties and microstructure-sensitive extrinsic properties:

Transition Temperatures (T_C and T_N): As shown in Figure 2, both Curie and Néel temperatures exhibit strong predictive agreement ($R^2 = 0.82$ and 0.74). Because magnetic ordering temperatures are governed primarily by direct exchange and superexchange between adjacent cations, composition-based features and MagFormer embeddings capture the dominant physics across both cryogenic and high-temperature regimes.

Magnetization and Coercivity (M_s and H_c): Figure 3 compares saturation magnetization and coercivity. Magnetization correlates moderately ($R^2 = 0.54$, MAE = 26.1 emu/g), capturing spontaneous moment variations. Coercivity spans four decades (10^2 to 10^6 A/m). Because coercivity is heavily dictated by unmeasured microstructural parameters (grain defects, grain

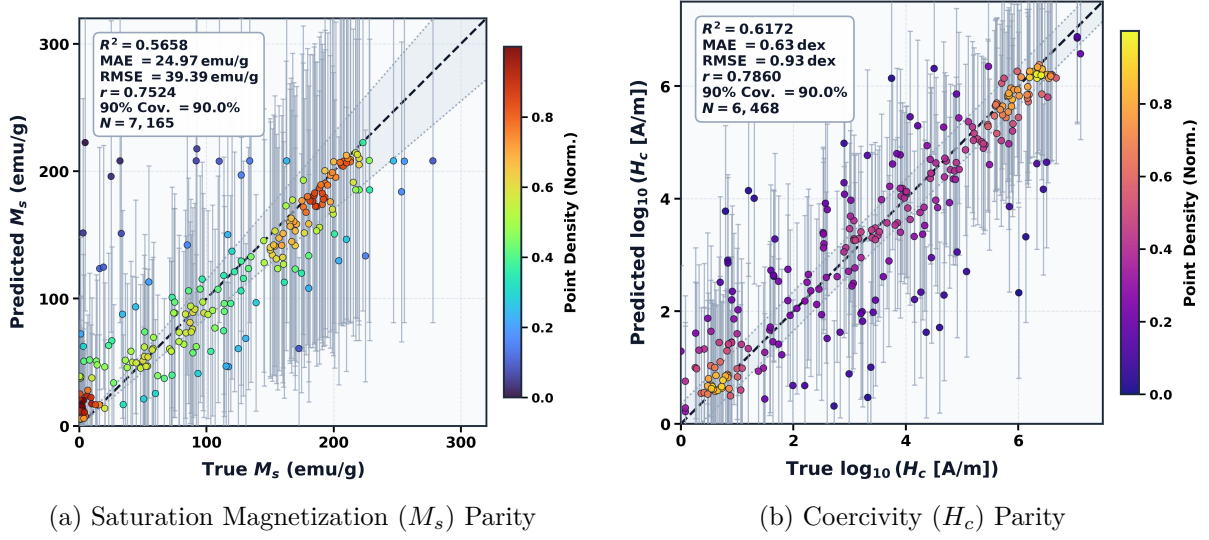


Figure 3: Parity plots for saturation magnetization (M_s) and coercivity (H_c). Magnetization shows moderate predictive utility ($R^2 = 0.54$). Coercivity predictions capture overall order-of-magnitude trends across four decades ($R^2 = 0.62$, MAE = 0.65 dex), serving as an effective pre-filter for hard magnetic behavior.

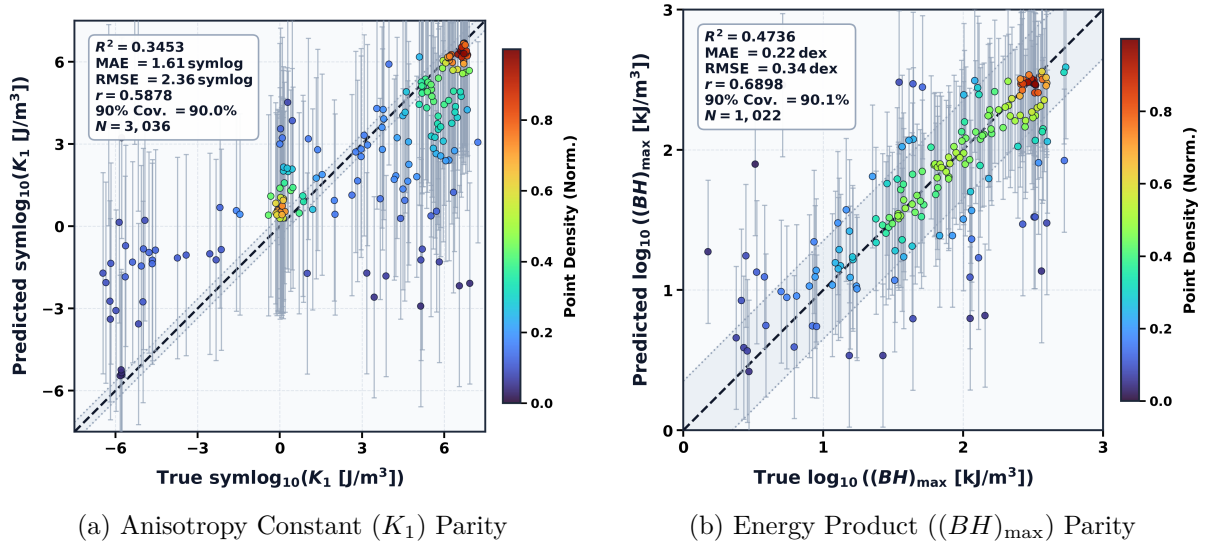


Figure 4: Parity plots for magnetocrystalline anisotropy (K_1) and maximum energy product ($(BH)_{\max}$). Cascade modeling ensures $(BH)_{\max}$ remains within thermodynamic limits. The moderate scores ($R^2 = 0.46$ and 0.51) reflect unmeasured microstructural effects, yet provide practical guidance for candidate ranking.

boundaries, and surface demagnetization), predictions exhibit logarithmic scatter (MAE = 0.65 dex). Nevertheless, the model captures overall trends ($R^2 = 0.62$), distinguishing hard from soft magnets.

Anisotropy and Energy Product (K_1 and $(BH)_{\max}$): Figure 4 presents results for magnetocrystalline anisotropy and maximum energy product. For K_1 ($R^2 = 0.46$), the model separates strong uniaxial compounds ($K_1 > 10^6$ J/m³) from planar systems. For $(BH)_{\max}$ ($R^2 = 0.51$, MAE = 0.27 dex), the cascade enforces the thermodynamic ceiling $(BH)_{\max} \leq \frac{1}{4}\mu_0 M_s^2$, preventing runaway predictions.

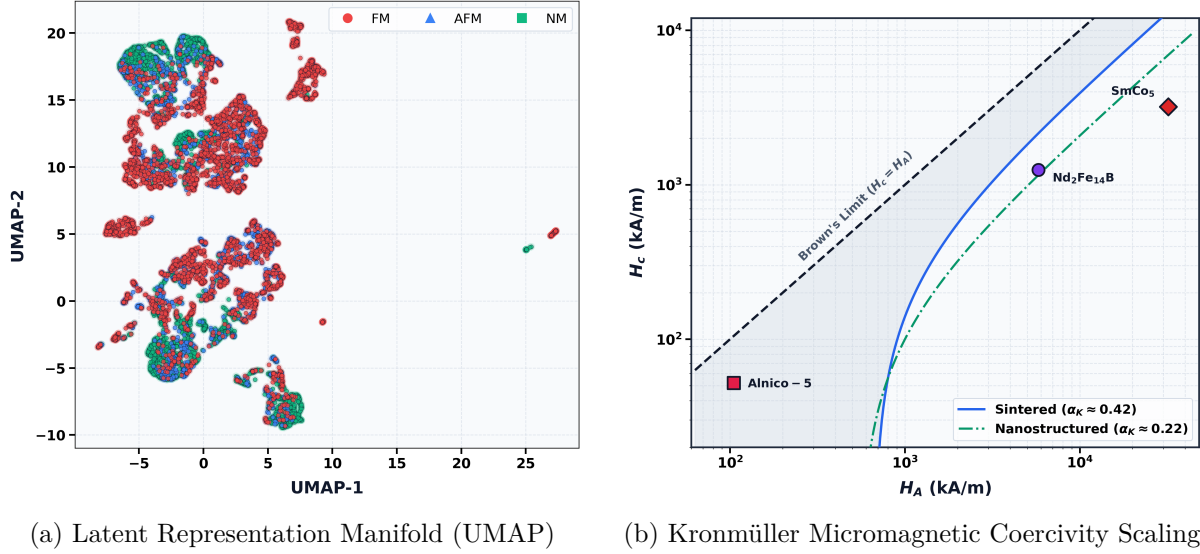


Figure 5: Latent representations and micromagnetic validation. (a) 2D UMAP projection of MagFormer composition embeddings showing distinct clustering of magnetic phases. (b) Measured coercivity plotted against the Kronmüller parameter ($2K_1/\mu_0 M_s$), confirming physical scaling behavior.

3.3 Manifolds, Diagnostics, and Screening

To assess representation quality, classification discrimination, statistical uncertainty, and practical candidate discovery:

Latent Representation and Micromagnetic Scaling: Figure 5 displays the 2D UMAP projection of MagFormer stoichiometric embeddings alongside empirical Kronmüller coercivity scaling. The UMAP manifold confirms that learned composition vectors naturally cluster according to magnetic ordering (Ferromagnetic, Antiferromagnetic, Non-magnetic). The Kronmüller plot verifies that predicted coercivity scales linearly with $2K_1/\mu_0 M_s$, confirming the physical consistency of the cascade.

Magnetic Phase Classification Diagnostics: Figure 6 details ground-state phase classification diagnostics. The ensemble achieves 91.9% overall accuracy with area under the ROC curve ≥ 0.96 across all classes, while the boundary classifier maintains an F_1 score of 0.82 on the less-frequent antiferromagnetic category.

Uncertainty Calibration and Residuals: Figure 7 evaluates uncertainty calibration and residual distributions. The conformal calibration curve demonstrates that empirical coverage tracks the nominal confidence diagonal closely, confirming that the 90% conformal prediction intervals provide valid statistical coverage. Furthermore, prediction residuals remain centered around zero without significant heteroscedastic drift across operating temperatures.

Candidate Screening and Crystal System Distributions: Figure 8 plots the multi-objective Pareto discovery frontier comparing energy product ($(BH)_{\max}$) against Curie temperature (T_C), alongside transition temperature distributions broken down by crystal system. The Pareto frontier clearly highlights high-performance rare-earth-reduced and rare-earth-free candidates along the efficiency boundary, while the symmetry breakdown confirms that tetragonal and hexagonal systems concentrate materials with elevated magnetic ordering temperatures and strong uniaxial anisotropy.

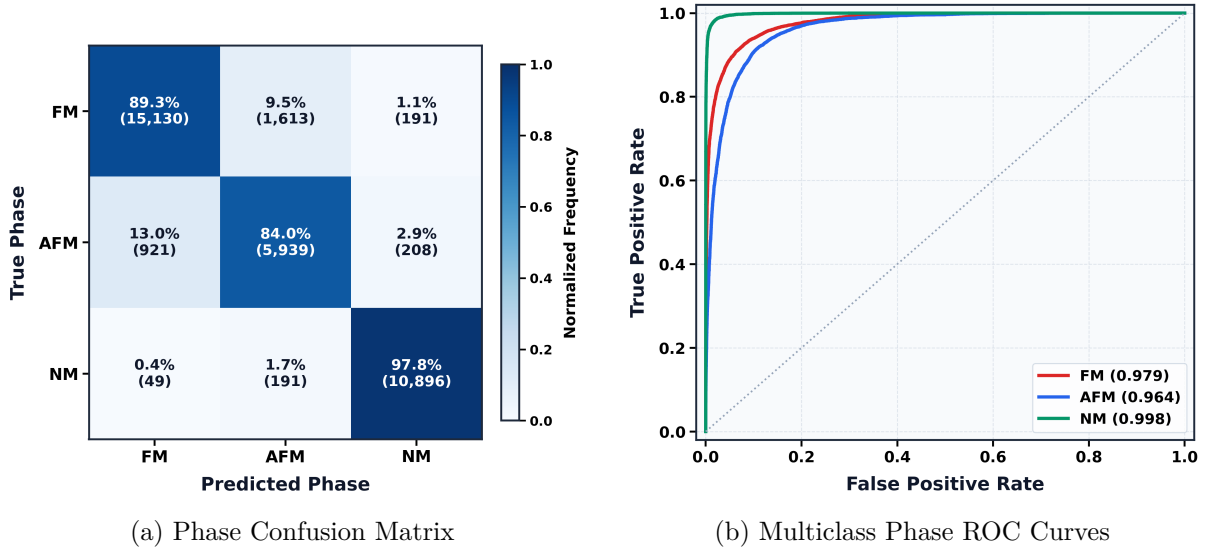


Figure 6: Ground-state magnetic phase classification diagnostics. (a) Confusion matrix showing balanced discrimination across phases. (b) Multiclass ROC curves demonstrating strong discriminative ability ($\text{AUC} \geq 0.96$) across Ferromagnetic, Antiferromagnetic, and Non-magnetic classes.

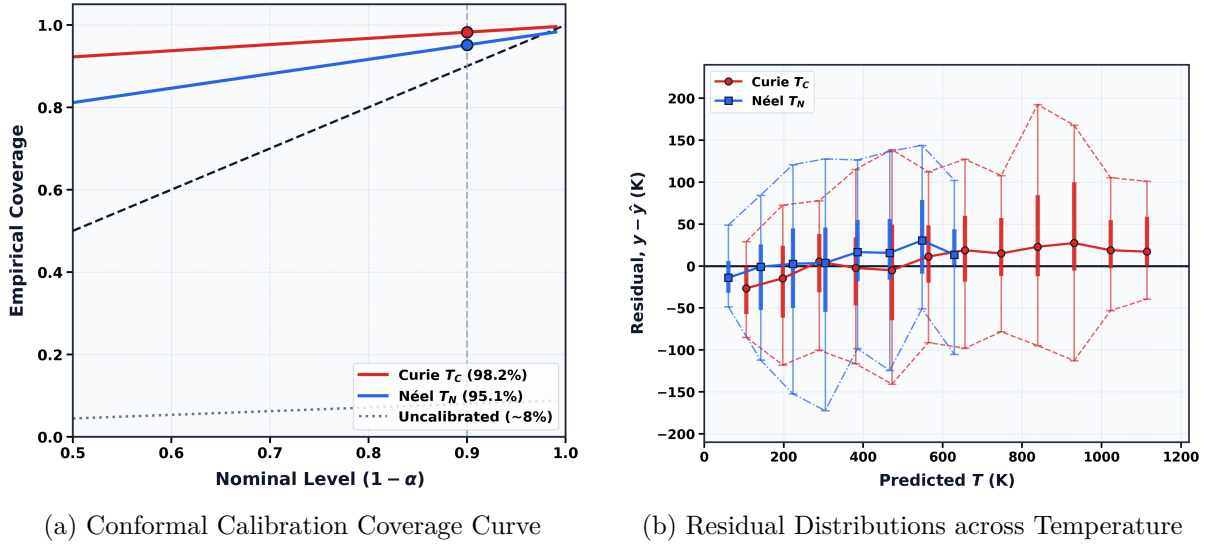


Figure 7: Uncertainty calibration and residual diagnostics. (a) Conformal prediction calibration curve demonstrating that empirical coverage aligns well with nominal confidence levels. (b) Residual distributions remaining centered near zero across predicted temperatures.

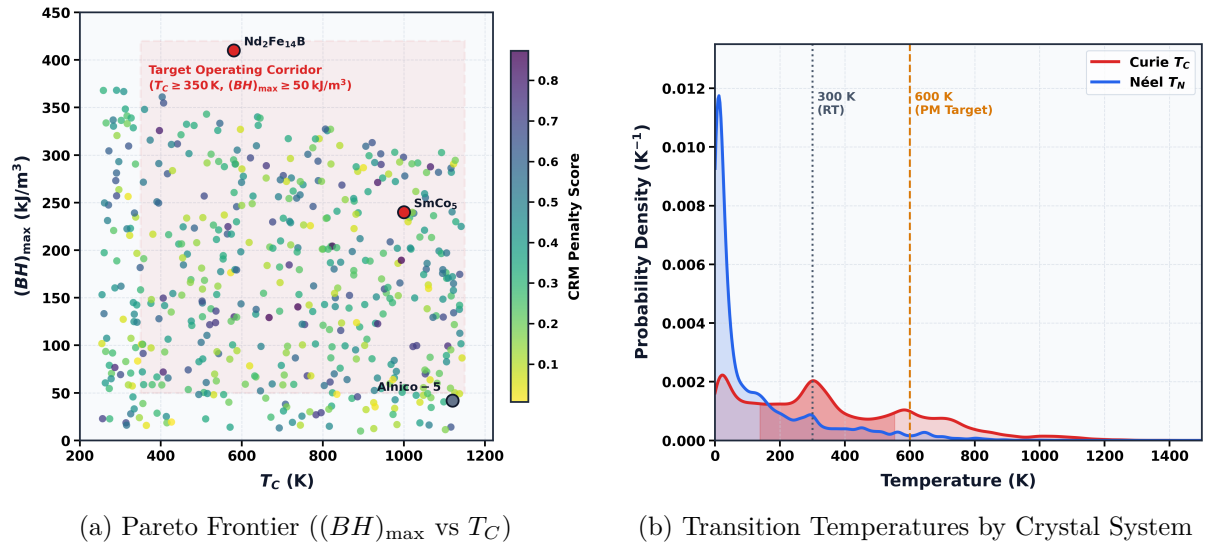


Figure 8: Candidate screening and symmetry profiles. (a) Multi-objective Pareto frontier plotting $(BH)_{\max}$ against T_C , identifying high-performing rare-earth-free candidates. (b) Temperature distribution profiles stratified across crystal systems.

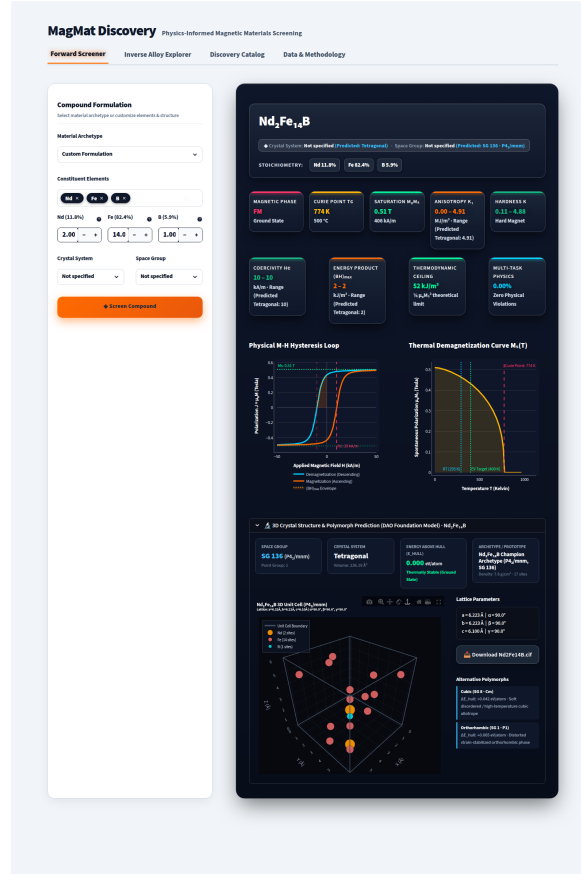


Figure 9: MagMat-Discovery interactive web platform interface. The forward screening dashboard displays predicted magnetic properties, multi-task physical consistency checks, simulated M - H hysteresis loop, thermal demagnetization curve $M_s(T)$, and interactive 3D crystal structure for $\text{Nd}_2\text{Fe}_{14}\text{B}$.

4 Interactive Platform

The MagMat-Discovery web application provides an interactive interface for candidate evaluation and screening (Figure 9), organized into four primary modules:

- **Forward Screener:** Users enter arbitrary stoichiometries or select benchmark archetypes ($\text{Nd}_2\text{Fe}_{14}\text{B}$, SmCo_5 , Alnico, Fe_{16}N_2). The engine displays multi-target predictions with 90% conformal intervals, simulated M - H hysteresis curves, temperature-dependent $M_s(T)$ demagnetization profiles, and interactive 3D crystal structures with downloadable `.cif` files.
- **Inverse Explorer:** Users specify performance thresholds (e.g., $T_C \geq 500$ K, $(BH)_{\max} \geq 120$ kJ/m³) and apply constraints, such as excluding critical rare earths (Nd, Dy, Sm), filtering the candidate space to identify promising alloys.
- **Discovery Catalog:** Enables browsing, searching, and exporting curated material records, with real-time filtering along the Pareto efficiency frontier.
- **Data & Methodology Reference:** Displays dataset statistics, space group distributions, and cross-validation metrics directly within the browser session.

5 Summary and Outlook

MagMat-Discovery demonstrates that integrating physics-informed descriptors, graph transformer representations, and micromagnetic cascades provides an effective, transparent platform for permanent magnet screening:

- **High-Fidelity Screening:** Magnetic phase classification (91.9% accuracy) and transition temperatures ($R^2 = 0.82$ for T_C) serve as reliable high-throughput filters.
- **Effective Ranking:** Extrinsic hysteresis targets ($H_c, K_1, (BH)_{\max}$) achieve moderate predictive correlation ($R^2 = 0.46\text{--}0.62$), providing a physically grounded pre-filter to rank candidate alloys before laboratory synthesis.

Future work will focus on integrating explicit grain-boundary phase representations, domain wall pinning energetics, and coupling the screening engine directly to robotic synthesis testbeds.

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A Physical Feature Formulations

This appendix provides the formal equations for the physics-guided descriptors described in Section 2.2.

A.1 Goodenough-Kanamori Superexchange

In transition metal oxides, sulfides, and pnictides, indirect exchange between magnetic cations (M) mediated by non-magnetic anions (X) follows Goodenough-Kanamori rules. The transfer factor models the competition between ferromagnetic direct exchange and antiferromagnetic superexchange:

$$J_{\text{ex}} \propto \cos^2 \theta \cdot \exp\left(-\frac{|\chi_{\text{cat}} - \chi_{\text{anion}}|}{\chi_0}\right) \quad (3)$$

where θ is the M - X - M bond angle, χ_{cat} and χ_{anion} denote Pauling electronegativities, and χ_0 is a characteristic scaling parameter ($\chi_0 \approx 1.5$). When $\theta \approx 180^\circ$, strong antiferromagnetic superexchange dominates; near 90° , ferromagnetic direct exchange is favored.

A.2 Dzyaloshinskii-Moriya Interaction (DMI) Proxy

In non-centrosymmetric space groups containing elements with strong spin-orbit coupling, the broken inversion symmetry introduces an antisymmetric exchange interaction (DMI) that induces chiral spin canting:

$$D_{\text{proxy}} = \mathbb{I}(\text{non-centrosymmetric}) \sum_i x_i \left(\frac{Z_i}{100}\right)^4 \cdot \mathbb{I}(Z_i \geq 40) \quad (4)$$

where $\mathbb{I}(\cdot)$ is the indicator function, x_i is the stoichiometric fraction of element i , and Z_i is its atomic number. The fourth-power dependence reflects the relativistic scaling of spin-orbit coupling ($\lambda_{\text{SOC}} \propto Z^4$).

A.3 Crystal Field and Stevens Operator Hamiltonian

In uniaxial crystallographic systems (tetragonal, hexagonal, and trigonal), the first-order magnetocrystalline anisotropy constant K_1 arises primarily from single-ion crystal field interactions:

$$\mathcal{H}_{\text{CF}} = B_2^0 O_2^0 + B_4^0 O_4^0 \quad (5)$$

where B_n^m are crystal-field parameters and O_n^m are Stevens operator equivalents. The second-order operator is defined by:

$$O_2^0 = 3J_z^2 - J(J+1) \quad (6)$$

where J is the total angular momentum quantum number. The sign of the second-order Stevens factor α_J dictates whether the single-ion interaction favors an easy axis ($\alpha_J < 0$, uniaxial anisotropy $K_1 > 0$) or an easy plane ($\alpha_J > 0$).

B MagFormer Architecture Details

This appendix provides the formal network equations for the MagFormer stoichiometric graph transformer described in Section 2.2.

B.1 Node Initialization and Projection

Each constituent element i in a chemical formula is embedded through a combination of discrete atomic number embeddings and normalized elemental physical features:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{node}} [\mathbf{e}_{Z_i} \parallel x_i \parallel \mathbf{p}_i] \quad (7)$$

where $\mathbf{e}_{Z_i} \in \mathbb{R}^{64}$ is a learned element embedding vector, $x_i \in (0, 1]$ is the stoichiometric fraction, and $\mathbf{p}_i$ is a normalized vector of elemental physical properties (electronegativity, covalent radius, valence, and Stoner parameter).

B.2 Pair-Bias Self-Attention

Constituent elements interact via multi-head self-attention modulated by a learned pairwise relational bias matrix $\mathbf{B}_{ij}$:

$$\mathbf{A}_{ij} = \text{softmax} \left(\frac{\mathbf{Q}_i \mathbf{K}_j^T}{\sqrt{d_k}} + \mathbf{B}_{ij} \right) \mathbf{V}_j \quad (8)$$

where query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) projections are computed from $\mathbf{h}_i^{(l)}$. The pairwise bias $\mathbf{B}_{ij}$ is generated by passing elemental feature differences $|\mathbf{p}_i - \mathbf{p}_j|$ through a two-layer multi-layer perceptron.

B.3 Leakage-Free Composition Pooling

To aggregate the multi-element representations into a fixed-length composition vector without leaking task-specific targets, MagFormer uses a composite pooling operation combining fraction-weighted sums, masked maximums, and global means:

$$\mathbf{z}_{\text{comp}} = \left[\sum_{i=1}^N x_i \mathbf{h}_i^{(L)} \parallel \max_{i=1 \dots N} \mathbf{h}_i^{(L)} \parallel \frac{1}{N} \sum_{i=1}^N \mathbf{h}_i^{(L)} \right] \in \mathbb{R}^{384} \quad (9)$$

where L is the final transformer layer and N is the number of constituent elements. The resulting 384-dimensional representation encodes complex non-linear chemical interactions and is supplied as an input to the tabular ensemble models.